Supporting Information: Probing Dimensional Crossover in SCAN Exchange with Exact Quasi-One-Dimensional Wires and Quasi-Two-Dimensional Sheets

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All results are fixed-density and exchange-only; no self-consistent calculation is involved. Every equation of the DC-SCAN construction is in Appendix A of the main text and is cited here by its main-text number. Section S1 derives the aniGTO integrals and density derivatives used for the exact references, and Section S8 lists the standalone scripts that produce every table and figure.

S1 aniGTO integrals and density derivatives used in this work

This section derives the quantities taken from the aniGTO anisotropic Gaussian package;[S1] the corresponding functions are copied verbatim into the scripts of Section S8.

S1.1 Anisotropic Cartesian Gaussians and the harmonic-trap basis

An aniGTO primitive has one exponent per Cartesian axis,

$$g_{\mathbf{l}}(\mathbf{r};\mathbf{a},\mathbf{A}) = \prod_{k=x,y,z} (x_k - A_k)^{l_k}\, e^{-a_k(x_k-A_k)^2}, \tag{S1}$$

and the normalized s function ($\mathbf{l}=0$) is $N_s\, g_0$ with

$$N_s = \prod_k \Big(\frac{2a_k}{\pi}\Big)^{1/4}. \tag{S2}$$

The harmonic-trap ("dot") basis for the trap frequencies $\boldsymbol{\omega} = (\omega_x, \omega_y, \omega_z)$ uses $a_k = \omega_k/2$, so its s function is the ground state of the anisotropic oscillator. Doubly occupied, it gives

$$n(\mathbf{r}) = 2N_s^2\, e^{-\sum_k \omega_k x_k^2} = n_A(\mathbf{r}), \qquad A = \mathrm{diag}(\boldsymbol{\omega}), \tag{S3}$$

which is the reference density of eqs A7 and A8 for $\boldsymbol{\omega} = (1, 1, \lambda^2)$ (sheet) and $(\lambda^2, \lambda^2, 1)$ (wire). The excited trap functions along z are the Hermite functions,

$$h_0(z) = \pi^{-1/4} e^{-z^2/2}, \quad h_1 = \sqrt{2}\, z\, h_0, \quad h_k = \sqrt{\frac{2}{k}}\, z\, h_{k-1} - \sqrt{\frac{k-1}{k}}\, h_{k-2}, \tag{S4}$$

so the $N = 4$ and $N = 6$ wires of Section S6 occupy $\phi_k = g(\rho)h_k(z)$, $k < N/2$, with $g^2(\rho) = (\lambda^2/\pi)e^{-\lambda^2\rho^2}$: the s, p_z and d_{zz} trap functions.

S1.2 One-dimensional building blocks

Along each axis the product of two Gaussians is one Gaussian (Gaussian product theorem),

$$e^{-a(x-A)^2} e^{-b(x-B)^2} = E_{AB}\, e^{-p(x-P)^2}, \qquad p = a + b, \quad P = \frac{aA + bB}{p}, \quad E_{AB} = e^{-\frac{ab}{p}(A-B)^2}, \tag{S5}$$

and every remaining integral reduces to the Gaussian moments

$$\int_{-\infty}^{\infty} x^m e^{-\beta x^2} dx = \begin{cases} \sqrt{\pi/\beta}\,(m-1)!!\,(2\beta)^{-m/2}, & m \text{ even}, \\ 0, & m \text{ odd}, \end{cases} \tag{S6}$$

with $(-1)!! = 1$. Products of shifted monomials are expanded binomially about the product centre U,

$$\int \prod_{i=1}^{4} (x - X_i)^{n_i} e^{-\beta(x-U)^2} dx = \sum_{j_1 \cdots j_4} \prod_i \binom{n_i}{j_i} (U - X_i)^{n_i - j_i} \int x^{j_1+j_2+j_3+j_4} e^{-\beta x^2} dx. \tag{S7}$$

S1.3 Two-electron repulsion integral

With the Gaussian transform of the Coulomb operator (eq A43 with integration variable u),

$$\frac{1}{r_{12}} = \frac{2}{\sqrt{\pi}} \int_0^{\infty} e^{-u^2 r_{12}^2} du, \tag{S8}$$

the six-dimensional integral over $\mathbf{r}_1, \mathbf{r}_2$ separates into three one-dimensional factors for every u:

$$(ab|cd) = \frac{2}{\sqrt{\pi}} E_{AB} E_{CD} \int_0^{\infty} du \prod_{k=x,y,z} I_k(u^2), \tag{S9}$$

$$I_k(u^2) = \iint (x_1 - A)^n (x_1 - B)^m (x_2 - C)^p (x_2 - D)^q \, e^{-p_{ab}(x_1-P)^2} e^{-p_{cd}(x_2-Q)^2} e^{-u^2(x_1-x_2)^2} dx_1 dx_2, \tag{S10}$$

where p_{ab}, P and p_{cd}, Q come from eq S5 for the pairs (a, b) and (c, d) along axis k. The x_1 integral is done first: completing the square,

$$e^{-p_{ab}(x_1-P)^2 - u^2(x_1-x_2)^2} = e^{-(p_{ab}+u^2)(x_1-X)^2} \, e^{-\xi(x_2-P)^2}, \qquad X = \frac{p_{ab}P + u^2 x_2}{p_{ab} + u^2}, \quad \xi = \frac{p_{ab} u^2}{p_{ab} + u^2}, \tag{S11}$$

and shifting $x_1 = y + X$ turns $x_1 - A$ into $y + \frac{u^2}{p_{ab}+u^2}(x_2 - A')$ with $A' = [A(p_{ab}+u^2) - p_{ab}P]/u^2$ (likewise B'). Expanding binomially and using eq S6 for y gives

$$I_k(u^2) = \sqrt{\frac{\pi}{p_{ab}+u^2}} \sum_{i\le n,\, j\le m}^{i+j \text{ even}} \binom{n}{i}\binom{m}{j} \frac{(i+j-1)!!}{[2(p_{ab}+u^2)]^{(i+j)/2}} \Big(\frac{u^2}{p_{ab}+u^2}\Big)^{n+m-i-j} e^{-\frac{p_{cd}\xi}{p_{cd}+\xi}(P-Q)^2}$$
$$\times \int (x_2 - A')^{n-i}(x_2 - B')^{m-j}(x_2 - C)^p (x_2 - D)^q e^{-(p_{cd}+\xi)(x_2-U)^2} dx_2, \qquad U = \frac{p_{cd}Q + \xi P}{p_{cd}+\xi}, \tag{S12}$$

and the last integral is eq S7 with $\beta = p_{cd} + \xi$. Eq S12 is the aniGTO kernel `integral3`; it is closed-form in u^2, so only the u integral of eq S9 is numerical.

Default quadrature. aniGTO maps $u \in [0, \infty)$ to $t = u/\sqrt{1+u^2} \in [0, 1)$, with $u^2 = t^2/(1-t^2)$ and $du = (1-t^2)^{-3/2} dt$:

$$(ab|cd) = \frac{2}{\sqrt{\pi}} E_{AB} E_{CD} \int_0^1 \frac{dt}{(1-t^2)^{3/2}} \prod_k I_k\Big(\frac{t^2}{1-t^2}\Big), \tag{S13}$$

evaluated by adaptive quadrature (`scipy.integrate.quad`, relative and absolute tolerances 10^{-16}). For an exponent ratio ρ_a the integrand peaks at $u^2 \sim \rho_a$, i.e. $1 - t \approx 1/(2\rho_a)$, which is the origin of the default-quadrature errors of Tables S1 and 4. The logarithmic variable $s = \ln u$ of eq 1 integrates the same integrand of eq S9 with $du = e^s ds$.

Self-repulsion of one s Gaussian. For four identical s functions at the origin with exponents a_k, $p_{ab} = p_{cd} = 2a_k$, $P = Q = 0$ and $E_{AB} = E_{CD} = 1$, so eq S12 has only the $i = j = 0$ term:

$$I_k(u^2) = \sqrt{\frac{\pi}{2a_k + u^2}} \sqrt{\frac{\pi}{2a_k + \xi}} = \frac{\pi}{2\sqrt{a_k(a_k + u^2)}}, \qquad \xi = \frac{2a_k u^2}{2a_k + u^2}, \tag{S14}$$

because $(2a_k+u^2)(2a_k+\xi)=4a_k(a_k+u^2)$. With $N_s^4=\prod_k 2a_k/\pi$ and $A_k=2a_k$ (eq S3),

$$(\phi\phi|\phi\phi)=N_s^4\,\frac{2}{\sqrt{\pi}}\int_0^\infty du\prod_k\frac{\pi}{2\sqrt{a_k(a_k+u^2)}}=\frac{2}{\sqrt{\pi}}\int_0^\infty\frac{\sqrt{\det A}}{\sqrt{\det(A+2u^2I)}}\,du, \tag{S15}$$

so $E_x=-(\phi\phi|\phi\phi)$ is exactly eq A44, and its closed forms are eqs A10 and A11.

S1.4 Density, gradient, Hessian and kinetic-energy density

For one primitive factor $f(x)=x^le^{-ax^2}$,

$$f'=\big(lx^{l-1}-2ax^{l+1}\big)e^{-ax^2},\qquad f''=\big(l(l-1)x^{l-2}-2a(2l+1)x^l+4a^2x^{l+2}\big)e^{-ax^2}, \tag{S16}$$

and the value, gradient and Hessian of $\phi=N\prod_k f_k(x_k)$ follow by the product rule. For a doubly occupied orbital

$$\begin{gathered}n=2\phi^2,\qquad \nabla n=4\phi\nabla\phi,\qquad \nabla\nabla n=4\big(\nabla\phi\,\nabla\phi^T+\phi\,\nabla\nabla\phi\big),\\ \tau=\tfrac{1}{2}\sum_i f_i|\nabla\phi_i|^2=|\nabla\phi|^2,\end{gathered} \tag{S17}$$

so $\tau_W=|\nabla n|^2/(8n)=\tau$ and $\alpha=0$ (eq A5). The curvature tensor eq A16 is formed from these grid values; for the s function it equals A (eq A17) at every point.

S1.5 Integration grids

Cylindrical logarithmic grid ($a_x=a_y=a_\perp$): with $u=2a_\perp\rho^2$ and $t=\sqrt{2a_z}\,z\ge 0$,

$$d^3r=2\pi\rho\,d\rho\,2\,dz=\frac{2\pi}{4a_\perp}\,\frac{2}{\sqrt{2a_z}}\,du\,dt, \tag{S18}$$

with 300–560 Gauss–Legendre nodes in $\ln u$ and $\ln t$, as in eq A33. *Gauss–Hermite product grid* of the aniGTO DFT module: per axis $x_i=\xi_i/\sqrt{2a_{\min}}$ and $w_i=w_i^{\mathrm{GH}}e^{\xi_i^2}/\sqrt{2a_{\min}}$ from the Gauss–Hermite rule $(\xi_i,w_i^{\mathrm{GH}})$, using the octant $x,y,z>0$ with weight 8.

S1.6 Exact exchange of several orbitals and the quasi-1D kernel

For the $N = 4$ and 6 wires the conventional exchange-energy density is

$$e_x^{\mathrm{HF}}(\mathbf{r}) = -\sum_{ij} \phi_i\phi_j(\mathbf{r})\, v_{ij}(\mathbf{r}), \qquad v_{ij}(\mathbf{r}) = \frac{2}{\sqrt{\pi}} \int_0^\infty dt\, \frac{\lambda^2}{\lambda^2+t^2}\, e^{-t^2 u/(\lambda^2+t^2)}\, Q_{ij}(z,t), \quad \text{(S19)}$$

with $u = \lambda^2\rho^2$ and $Q_{ij}(z,t) = \int h_i h_j(z') e^{-t^2(z-z')^2} dz'$, which eq S6 gives in closed form: the transverse Gaussian integral of eq S8 is done analytically and only t is numerical. Integrating the transverse Gaussians of both electrons first gives the quasi-1D (Kantorovich) form of the exact exchange,

$$E_x = -\sum_{ij} \iint h_i h_j(z)\, h_i h_j(z')\, w_b(z-z')\, dz\, dz', \qquad w_b(z) = \frac{\sqrt{\pi}}{2b}\, \mathrm{erfcx}\Big(\frac{|z|}{2b}\Big), \quad b = \frac{1}{\sqrt{2}\lambda}, \quad \text{(S20)}$$

which for $N = 2$ reduces to $E_x = -\frac{1}{4}\iint n(z)n(z')w_b(z-z')\, dz\, dz'$ and reproduces eq A11 to 10^{-15} Ha.

S2 Numerical protocol and checks

The exchange integral eq A33 is evaluated with s, D and Z in closed form on the (u,t) grid, with 400 Gauss–Legendre nodes per variable in $\ln u$ and $\ln t$, $u \in [e^{-32}, 26]$, $t \in [e^{-32}, 8]$; the curves of Figures S1 and S2 use 39 values of λ and 300 nodes. The constant M_{2D} of eq A28 is reproduced to machine precision, and all eighteen exact, SCAN and DC-SCAN entries of the wire table (Table 1) agree with an independent evaluation to the six printed decimals. Table S1 cross-checks the exact reference three ways: the closed form eq A11, an independent quadrature of eq A44, and the aniGTO primitive repulsion integral $-(\phi\phi|\phi\phi)$ with exponents $(\lambda^2/2, \lambda^2/2, 1/2)$. For $\lambda \le 10^3$ the aniGTO value agrees to 3×10^{-12} Ha. At $\lambda = 10^4$ the exponent ratio is 10^8 and the integrand of the generalized-Boys transform is confined to $1-t \approx 10^{-8}$; the default adaptive quadrature then misses by 6.7×10^{-5} Ha, and the same

integrand with breakpoints at $t = 1-10^{-k}$ ($k = 1, \ldots, 9$) agrees to 8×10^{-11} Ha. Raising the exchange quadrature from 400 to 560 nodes changes DC-SCAN by at most 4.6×10^{-8} Ha.

Table S1: Exact exchange of the Gaussian wire and numerical checks (hartree). "aniGTO" is the aniGTO repulsion integral $-(\phi\phi|\phi\phi)$ (eqs S14–S15) evaluated with the default adaptive quadrature in t (eq S13); "aniGTO (fine)" the same integrand of eq S13 with breakpoints at $t = 1-10^{-k}$, $k = 1, \ldots, 9$. The last column is the change of E_x^{DC} from 400 to 560 quadrature nodes. All computed values are listed first (exact closed form, eq A44, the two aniGTO evaluations and E_x^{DC} on both quadratures); the differences follow at the end of each row.

	computed values						differences			
λ	E_x^{exact}, eq A11	eq A44	aniGTO, eq S13	aniGTO (fine), eq S13	E_x^{DC} (400)	E_x^{DC} (560)	eq A44−exact	aniGTO−exact	aniGTO (fine)−exact	$\Delta E_{400\to560}^{\mathrm{DC}}$
1	-0.7978845608	-0.7978845608	-0.7978845608	-0.7978845608	-0.7950576394	-0.7950576394	-6.1×10^{-15}	2.2×10^{-16}	0	2.0×10^{-14}
2	-1.2133366626	-1.2133366626	-1.2133366626	-1.2133366626	-1.0601354096	-1.0601353633	1.5×10^{-14}	0	4.4×10^{-16}	4.6×10^{-8}
10	-2.4002778396	-2.4002778396	-2.4002778396	-2.4002778396	-2.1268627834	-2.1268628154	2.9×10^{-13}	0	8.9×10^{-16}	-3.2×10^{-8}
100	-4.2276370643	-4.2276370643	-4.2276370643	-4.2276370643	-4.1938784469	-4.1938784469	8.7×10^{-13}	-3.2×10^{-14}	-1.1×10^{-14}	-2.1×10^{-11}
1,000	-6.0646455535	-6.0646455535	-6.0646455535	-6.0646455536	-6.1707886855	-6.1707886855	1.4×10^{-12}	-3.0×10^{-12}	-9.1×10^{-11}	1.1×10^{-13}
10,000	-7.9018398538	-7.9018398538	-7.9019067908	-7.9018398539	-7.9904378764	-7.9904378764	-1.3×10^{-11}	-6.7×10^{-5}	-8.0×10^{-11}	1.9×10^{-13}

Table S2: Local slopes on the Gaussian wire at the geometric midpoints of successive $\lambda = 10^2, 10^{2.5}, \ldots, 10^5$. The exact log coefficient is $-\sqrt{2/\pi} = -0.797885$; SCAN's exponent is predicted to approach 1/2 (eq A15). The last column is DC-SCAN−exact at the lower end of each interval (hartree).

λ_{mid}	$dE_x^{\mathrm{exact}}/d\ln\lambda$	$dE_x^{\mathrm{DC}}/d\ln\lambda$	$d\ln\lvert E_x^{\mathrm{SCAN}}\rvert/d\ln\lambda$	DC−exact
1.8×10^2	-0.797739	-0.885908	0.6406	+0.0338
5.6×10^2	-0.797866	-0.831214	0.6314	-0.0678
1.8×10^3	-0.797882	-0.796522	0.6211	-0.1061
5.6×10^3	-0.797884	-0.784005	0.6102	-0.1046
1.8×10^4	-0.797885	-0.783530	0.5992	-0.0886
5.6×10^4	-0.797885	-0.787048	0.5883	-0.0721

Table S3: Asymptotic DC-SCAN offset on the Gaussian wire (hartree), from the same quadrature with enlarged ranges. The last column is the DC-SCAN slope $dE_x/d\ln\lambda$ over the interval ending at λ; the exact coefficient is $-\sqrt{2/\pi} = -0.797885$.

λ	E_x^{exact}, eq A11	E_x^{DC}	DC−exact	$dE_x^{\mathrm{DC}}/d\ln\lambda$
10^3	-6.064646	-6.170789	-0.106143	
10^4	-7.901840	-7.990438	-0.088598	-0.79026
10^5	-9.739037	-9.798632	-0.059595	-0.78529
10^6	-11.576234	-11.622719	-0.046485	-0.79219
10^8	-15.250628	-15.291421	-0.040793	-0.79665
10^{10}	-18.925022	-18.965370	-0.040348	-0.79779
10^{12}	-22.599417	-22.639735	-0.040319	-0.79788

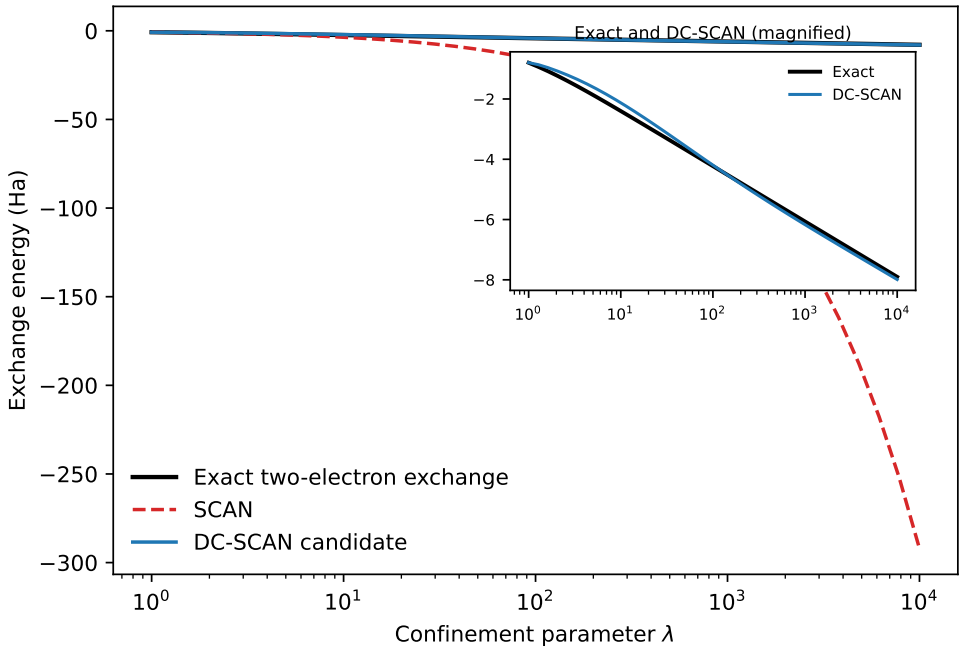


Figure S1: Exchange energy of the two-electron Gaussian wire against the confinement parameter λ: exact (eq A11), SCAN and DC-SCAN (eq A20). SCAN diverges as a power of λ; the inset magnifies the exact and DC-SCAN curves and shows the residual offset of DC-SCAN.

S3 Evaluation inside aniGTO

The wire orbital $\phi = \sqrt{n_\lambda/2}$ is exactly the s function of the aniGTO harmonic-trap basis with $\boldsymbol{\omega} = (\lambda^2, \lambda^2, 1)$, whose exponents are $(\lambda^2/2, \lambda^2/2, 1/2)$. The aniGTO columns of Table 1 are computed with aniGTO objects only:

- the exact exchange is the aniGTO repulsion integral $-(\phi\phi|\phi\phi)$ of the trap molecule;
- n, ∇n, the Hessian $\nabla\nabla n$, $\nabla^2 n$ and $\tau = \frac{1}{2}\sum_i f_i|\nabla\phi_i|^2$ are evaluated from analytic first and second derivatives of the aniGTO Cartesian Gaussians;
- LDA, PBE, SCAN and r^{2}SCAN exchange are taken from libxc through PySCF, $F_x = e_x/e_x^{\mathrm{LDA}}$ pointwise, and D, Z, s and α (eqs A4, A5 and A18) are formed on the grid from the aniGTO Hessian rather than from the closed form $C = A$.

On these one-orbital densities $|\alpha| \leq 10^{-8}$, and libxc SCAN and r^{2}SCAN agree with the closed-form $\alpha = 0$ branch (eq A6) to the printed digits. As a second exact route, the quasi-1D Kantorovich reduction gives $E_x = -\frac{1}{4}\iint n(z)n(z')\,w_b(z-z')\,dz\,dz'$ with $w_b(z) = \sqrt{\pi}/(2b)\,\mathrm{erfcx}(|z|/2b)$ and $b = 1/(\sqrt{2}\lambda)$. It is exact here because both electrons share the transverse Gaussian, and it reproduces eq A11 to 10^{-15} Ha for all λ.

Table S4 compares two grids. The anisotropic Gauss–Hermite product grid of the aniGTO DFT module, scaled per axis by the smallest exponent, converges SCAN to $\leq 3 \times 10^{-3}$ Ha at every λ. DC-SCAN, however, is switched on only in the region $\lambda^{-2/3} \ll \lambda^2\rho^2 \ll 1$ that carries the wire logarithm, and that region spans many decades in ρ. A product grid uniform in ρ cannot resolve it, so for $\lambda \geq 10^3$ DC-SCAN on that grid is not converged. A cylindrical grid, logarithmic in $u = \lambda^2\rho^2$ and $t = z$ (the coordinates of eq A33) and evaluated at points of the aniGTO basis functions, converges every entry to 10^{-7} Ha. A Hessian-dependent functional thus needs a logarithmic radial grid around the confinement axis before it can be used self-consistently in aniGTO.

Table S4: aniGTO evaluation of the wire exchange (hartree) on the Gauss–Hermite grid of the aniGTO DFT module (nodes per axis) and on the cylindrical logarithmic grid (Gauss–Legendre nodes per variable). Computed values first: N is the integrated electron number and the next seven columns are the exchange energies; then the differences: LDA, SCAN and DC-SCAN minus the reference quadrature of Table 1, and the closed-form SCAN ($\alpha = 0$ branch) evaluated on the same grid minus libxc SCAN.

				computed values: aniGTO fields (eq S17), grid (eq S18), libxc							differences			
λ	grid	nodes	N	LDA	PBE	SCAN	r^{2}SCAN	DC-SCAN	DC-PBE	DC-r^{2}SCAN	LDA−ref	SCAN−ref	DC-SCAN−ref	SCAN^{cf}−SCAN
10	Gauss–Hermite	80	2.00000000	-3.165498	-3.865468	-3.649731	-3.649731	-2.126716	-1.976367	-2.126716	+5.1e-14	+3.3e-07	+1.5e-04	-4.4e-16
10	Gauss–Hermite	160	2.00000000	-3.165498	-3.865468	-3.649731	-3.649731	-2.126888	-1.976753	-2.126888	+5.1e-14	+4.4e-08	-2.5e-05	+0.0e+00
10	Gauss–Hermite	240	2.00000000	-3.165498	-3.865468	-3.649731	-3.649731	-2.126817	-1.976674	-2.126817	+5.2e-14	+6.5e-09	+4.6e-05	+0.0e+00
10	cylindrical log	300	2.00000000	-3.165498	-3.865468	-3.649731	-3.649731	-2.126863	-1.976721	-2.126863	+9.4e-14	+1.1e-13	+7.6e-08	+4.4e-16
10	cylindrical log	400	2.00000000	-3.165498	-3.865468	-3.649731	-3.649731	-2.126863	-1.976722	-2.126863	+4.4e-16	+1.8e-15	+1.3e-15	+0.0e+00
10	cylindrical log	560	2.00000000	-3.165498	-3.865468	-3.649731	-3.649731	-2.126863	-1.976722	-2.126863	+8.1e-14	+9.4e-14	-3.2e-08	+0.0e+00
100	Gauss–Hermite	80	2.00000000	-14.692939	-20.817613	-16.362346	-16.362346	-4.209902	-4.014640	-4.209902	+2.3e-13	-4.7e-05	-1.6e-02	+0.0e+00
100	Gauss–Hermite	160	2.00000000	-14.692939	-20.817581	-16.362302	-16.362302	-4.186642	-3.984922	-4.186642	+2.3e-13	-3.6e-06	+7.2e-03	-3.6e-15
100	Gauss–Hermite	240	2.00000000	-14.692939	-20.817578	-16.362299	-16.362299	-4.195736	-3.994080	-4.195736	+2.3e-13	-1.5e-07	-1.9e-03	+0.0e+00
100	cylindrical log	300	2.00000000	-14.692939	-20.817577	-16.362299	-16.362299	-4.193878	-3.992240	-4.193878	+4.3e-13	+4.8e-13	-1.3e-10	+0.0e+00
100	cylindrical log	400	2.00000000	-14.692939	-20.817577	-16.362299	-16.362299	-4.193878	-3.992240	-4.193878	+0.0e+00	-3.6e-15	-1.8e-15	+0.0e+00
100	cylindrical log	560	2.00000000	-14.692939	-20.817577	-16.362299	-16.362299	-4.193878	-3.992240	-4.193878	+3.7e-13	+4.1e-13	-2.1e-11	+0.0e+00
1,000	Gauss–Hermite	80	2.00000000	-68.198582	-109.476193	-70.764359	-70.764359	-7.158316	-7.024959	-7.158316	+1.1e-12	-1.4e-03	-9.9e-01	-2.8e-14
1,000	Gauss–Hermite	160	2.00000000	-68.198582	-109.470731	-70.763363	-70.763363	-6.161318	-5.916625	-6.161318	+1.1e-12	-4.0e-04	+9.5e-03	-1.4e-14
1,000	Gauss–Hermite	240	2.00000000	-68.198582	-109.470303	-70.763121	-70.763121	-6.022794	-5.821126	-6.022794	+1.1e-12	-1.6e-04	+1.5e-01	-2.8e-14
1,000	cylindrical log	300	2.00000000	-68.198582	-109.470149	-70.762964	-70.762964	-6.170789	-5.965216	-6.170789	+2.0e-12	+2.1e-12	+9.9e-14	-1.4e-14
1,000	cylindrical log	400	2.00000000	-68.198582	-109.470149	-70.762964	-70.762964	-6.170789	-5.965216	-6.170789	-2.8e-14	+1.4e-14	-2.7e-15	-2.8e-14
1,000	cylindrical log	560	2.00000000	-68.198582	-109.470149	-70.762964	-70.762964	-6.170789	-5.965216	-6.170789	+1.7e-12	+1.8e-12	+1.0e-13	-2.8e-14
10,000	Gauss–Hermite	80	2.00000000	-316.549776	-546.118619	-292.057294	-292.057294	-4.008572	-4.061073	-4.008572	+5.0e-12	+7.5e-03	+4.0e+00	-5.7e-13
10,000	Gauss–Hermite	160	2.00000000	-316.549776	-545.687062	-292.067275	-292.067275	-5.768257	-5.915872	-5.768257	+4.9e-12	-2.5e-03	+2.2e+00	-5.1e-13
10,000	Gauss–Hermite	240	2.00000000	-316.549776	-545.622921	-292.067066	-292.067066	-7.896087	-7.982759	-7.896087	+5.1e-12	-2.3e-03	+9.4e-02	-5.1e-13
10,000	cylindrical log	300	2.00000000	-316.549776	-545.596537	-292.064785	-292.064785	-7.990438	-7.788276	-7.990438	+9.3e-12	+9.2e-12	+3.2e-13	-4.5e-13
10,000	cylindrical log	400	2.00000000	-316.549776	-545.596537	-292.064785	-292.064785	-7.990438	-7.788276	-7.990438	+5.7e-14	+4.5e-13	+8.9e-16	-4.0e-13
10,000	cylindrical log	560	2.00000000	-316.549776	-545.596537	-292.064785	-292.064785	-7.990438	-7.788276	-7.990438	+8.1e-12	+7.7e-12	+1.9e-13	-4.0e-13

S4 Gaussian sheet

The sheet table of the main text (Table 2) uses eq A33 with $k = 1$ and the same grid. All eighteen exact, SCAN and DC-SCAN entries agree with an independent evaluation to the six printed decimals. Raising the quadrature from 400 to 560 nodes changes DC-SCAN by at most 7.6×10^{-7} Ha (at $\lambda = 2$) and SCAN by 1.5×10^{-13} Ha; widening the ranges to $u \leq 34$, $t \leq 10$, lower limits e^{-40}, changes DC-SCAN by at most 2.7×10^{-7} Ha. Table S5 repeats the exact-exchange checks of Table S1 for the sheet, with aniGTO exponents $(1/2, 1/2, \lambda^2/2)$. Table S6 follows the three energies toward their $\lambda \to \infty$ limits. The exact energy approaches $-\sqrt{\pi/2}$ (eq A12) as $\sqrt{2/\pi}\,\lambda^{-1}$, DC-SCAN$-$exact decays as $\lambda^{-1/3}$ (local exponent -0.333 over the last decade), and SCAN approaches its own limit $-h_x^0 a_1 M_{2D} = -11.145141$ Ha only as about $\lambda^{-0.23}$; that limit is reached only where $s \gg a_1^2 \approx 24.5$ over most of the sheet.

Table S5: Exact exchange of the Gaussian sheet and numerical checks (hartree), as in Table S1; closed form eq A10.

	computed values						differences			
λ	$E_x^{\rm exact}$, eq A10	eq A44	aniGTO, eq S13	aniGTO (fine), eq S13	$E_x^{\rm DC}$ (400)	$E_x^{\rm DC}$ (560)	eq A44$-$exact	aniGTO$-$exact	aniGTO (fine)$-$exact	$\Delta E_{400\to560}^{\rm DC}$
1	-0.7978845608	-0.7978845608	-0.7978845608	-0.7978845608	-0.7950576394	-0.7950576394	-6.1×10^{-15}	2.2×10^{-16}	0	2.0×10^{-14}
2	-0.9648016727	-0.9648016727	-0.9648016727	-0.9648016727	-0.8693771541	-0.8693763930	-2.2×10^{-16}	1.1×10^{-16}	1.1×10^{-16}	7.6×10^{-7}
10	-1.1793034311	-1.1793034311	-1.1793034311	-1.1793034311	-1.0829953753	-1.0829953760	3.6×10^{-14}	4.4×10^{-16}	2.2×10^{-16}	-6.6×10^{-10}
100	-1.2453974301	-1.2453974301	-1.2453974301	-1.2453974301	-1.1847788041	-1.1847788020	5.5×10^{-14}	0	-1.1×10^{-15}	2.2×10^{-9}
1,000	-1.2525168789	-1.2525168789	-1.2525168789	-1.2525168789	-1.2214079455	-1.2214079461	5.8×10^{-14}	-1.3×10^{-15}	-6.0×10^{-15}	-5.0×10^{-10}
10,000	-1.2532343551	-1.2532343551	-1.2533141435	-1.2532343551	-1.2385060219	-1.2385060219	5.7×10^{-14}	-8.0×10^{-5}	-6.2×10^{-14}	6.9×10^{-12}

Table S6: Approach of the Gaussian sheet to the $\lambda \to \infty$ limits (hartree). Computed values first ($E_x^{\rm exact}$, DC-SCAN, SCAN), then the differences: exact minus $-\sqrt{\pi/2}$, DC-SCAN minus exact, and SCAN minus its limit $-h_x^0 a_1 M_{2D} = -11.145141$.

	computed values			differences		
λ	$E_x^{\rm exact}$, eq A10	$E_x^{\rm DC}$, eq A20	$E_x^{\rm SCAN}$	$E_x^{\rm exact} + \sqrt{\pi/2}$	DC$-$exact	$E_x^{\rm SCAN} - E_\infty^{\rm SCAN}$
10^2	-1.245397	-1.184779	-3.179886	7.9×10^{-3}	+0.060619	+7.9653
$10^{2.5}$	-1.250797	-1.206554	-4.128303	2.5×10^{-3}	+0.044244	+7.0168
10^3	-1.252517	-1.221408	-5.130958	8.0×10^{-4}	+0.031109	+6.0142
$10^{3.5}$	-1.253062	-1.231573	-6.121554	2.5×10^{-4}	+0.021489	+5.0236
10^4	-1.253234	-1.238506	-7.045312	8.0×10^{-5}	+0.014728	+4.0998
$10^{4.5}$	-1.253289	-1.243228	-7.866517	2.5×10^{-5}	+0.010061	+3.2786
10^5	-1.253306	-1.246444	-8.568654	8.0×10^{-6}	+0.006862	+2.5765
$10^{5.5}$	-1.253312	-1.248634	-9.150344	2.5×10^{-6}	+0.004677	+1.9948
10^6	-1.253313	-1.250126	-9.620105	8.0×10^{-7}	+0.003187	+1.5250

S4.1 Validation of the sheet construction against the literature

In the $\lambda \to \infty$ (2D) limit of a one-orbital density, s diverges, and any enhancement factor with $F_x \to A\, s^{-1/2}$ gives the finite exchange energy

$$E_x \to -C_X A\, K \int n^2\, |\partial_z n|^{-1/2}\, d^3r, \qquad K = \left[2(3\pi^2)^{1/3}\right]^{1/2}, \tag{S21}$$

because $n^{4/3} s^{-1/2} = K\, n^2 |\partial_z n|^{-1/2}$. Fixing A so that eq S21 equals the exact 2D limit gives the DC-SCAN sheet coefficient A_{2D} (eqs A28–A29); setting $A = h_x^0 a_1$ gives the SCAN limit. We applied the same formula, unchanged, to the infinite-barrier slab

$$n(z) = n_{2D}\, \frac{2}{L} \sin^2 \frac{\pi z}{L}, \quad n_{2D} = \frac{1}{\pi r_s^2}, \qquad \int n^2 |n'|^{-1/2} dz = \frac{(2n_{2D})^{3/2}}{\sqrt{2}\,\pi^{3/2}}\, B\left(\tfrac{9}{4}, \tfrac{1}{4}\right), \tag{S22}$$

with B the Beta function. It reproduces Constantin's coefficient a_x for the exact 2D limit of the slab and the SCAN 2D limit of Kaplan, Wagle and Perdew, both obtained independently (Table S7); a_x is independent of r_s^{2D} to all printed digits. The remaining 0.9% difference to the numerical SCAN value of ref S2 is that of their quadrature at $L/r_s^{2D} = 10^{-10}$ against our exact limit; their own analytic estimate agrees to 0.02%.

Table S7: The sheet-limit construction of DC-SCAN (eq S21) applied to the infinite-barrier slab of refs S2,S3 and to the Gaussian sheet of this work. aniGTO: the same quantity from the aniGTO exact exchange of the Gaussian sheet at $\lambda = 10^8$ (eqs S14–S15 in the logarithmic variable of eq 1), -1.253314129337 Ha, equal to the closed form eq A10 to 12 digits; "–": not applicable (the $\sin^2$ slab is not a Gaussian density). Computed values first, then the literature or reference value, its source, and the relative differences at the end.

	this work				difference to reference	
quantity	eq S21	aniGTO, eq S15	reference	source	eq S21	aniGTO
slab: exact 2D E_x/N (Ha), $r_s^{2D} = 4$	-0.150053	–	-0.1501	ref S2	+0.03%	–
slab: a_x giving the exact 2D limit	0.521748	–	0.5217	ref S3, eq 12	+0.01%	–
slab: SCAN 2D limit E_x/N (Ha), analytic	-1.670599	–	-1.6710	ref S2, App. A	+0.02%	–
slab: SCAN 2D limit E_x/N (Ha), numerical	-1.670599	–	-1.6550	ref S2, numerical	−0.94%	–
slab: SCAN/exact	11.133411	–	11.0260	ref S2	+0.97%	–
Gaussian sheet: exact 2D limit E_x (Ha)	-1.253314	-1.253314	-1.2533	eq A12	+0.00%	+0.00%
Gaussian sheet: A_{2D}	0.653226	0.653226	0.6532	eq A29	+0.00%	−0.00%
Gaussian sheet: SCAN limit (Ha)	-11.145141	–	-11.1451	eq A28, Table 2	+0.00%	–
Gaussian sheet: SCAN/exact	8.892536	8.892536	8.8900	main text	+0.03%	+0.03%

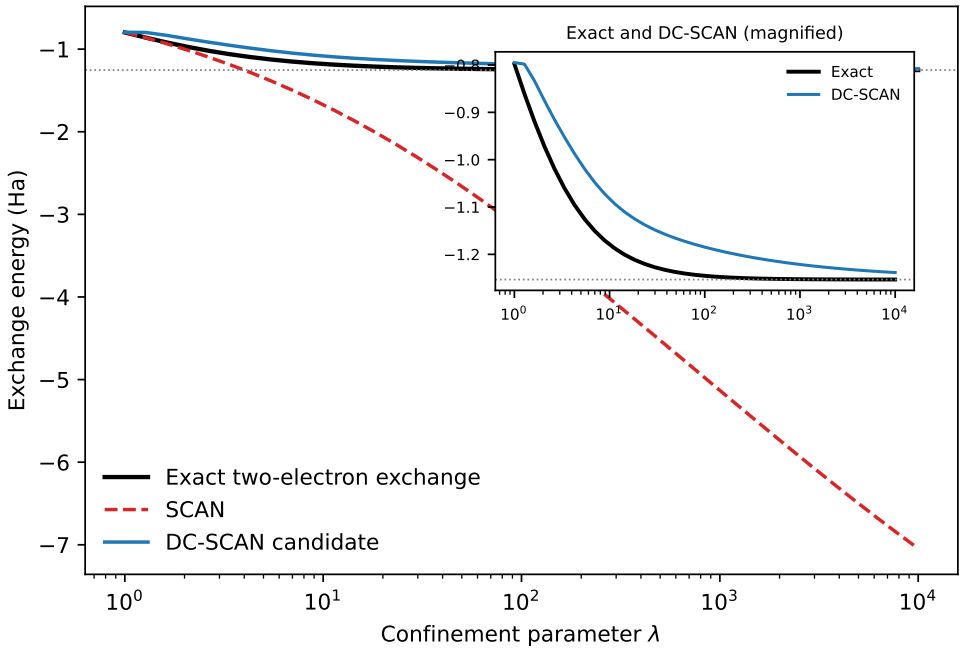


Figure S2: Exchange energy of the two-electron Gaussian sheet against the confinement parameter λ: exact, SCAN and DC-SCAN (eq A20). The dotted line is the exact limit $-\sqrt{\pi/2}$; the inset magnifies the exact and DC-SCAN curves.

S5 PBE base and refit of the activation scales

Replacing F_x^{SCAN} by the PBE enhancement factor in eq A20 (DC-PBE) keeps the sheet limit and the wire log coefficient, since both are set by the dimensional branches. PBE alone, whose F_x saturates at 1.804, grows like LDA. The activation function eq A25 has two scales that the construction sets to one; it is generalized here to

$$\Omega = \frac{(Z/Z_0)^3}{1+(Z/Z_0)^3}\,\frac{(s/s_0)^8}{1+(s/s_0)^8}. \tag{S23}$$

Any $s_0, Z_0 > 0$ preserve the uniform-gas limit, the Gaussian sheet limit and the wire log coefficient, because A_{2D} and A_{1D} are not changed. Fitting only s_0 and Z_0 to the wire errors at $\lambda = 2$–10^4 and using the sheet family as an independent test set gives Table S8. The fit selects $s_0 \approx 1$ and $Z_0 \approx 2.3$ (SCAN) or 2.6 (PBE), lowering the wire RMS error by a factor of

5–6. The sheet errors, which were not fitted, change by at most 0.08 Ha, and the sheet RMS improves. This is a two-parameter, one-family fixed-density fit, reported as a diagnostic of where the remaining finite-width error comes from, not as a validated parametrization.

Table S8: Exchange energies E_x and errors $E_x - E_x^{\text{exact}}$ (hartree) of the base functionals, DC-SCAN/DC-PBE with unit activation scales, and the wire-fitted scales (s_0, Z_0), on the wire (fit set) and the sheet (test set). The exact closed-form values come first, then for every variant the computed E_x, and the errors and their RMS at the end of the row.

		E_x (Ha)					$E_x - E_x^{\text{exact}}$ (Ha)					
variant	family	$\lambda = 2$	10	100	10^3	10^4	$\lambda = 2$	10	100	10^3	10^4	RMS
exact, eq A11	wire	-1.213337	-2.400278	-4.227637	-6.064646	-7.901840						
exact, eq A10	sheet	-0.964802	-1.179303	-1.245397	-1.252517	-1.253234						
SCAN alone	wire	-1.260632	-3.649731	-16.362299	-70.762964	-292.064785	-0.0473	-1.2495	-12.1347	-64.6983	-284.1629	130.4479
SCAN alone	sheet	-1.000228	-1.672690	-3.179886	-5.130958	-7.045312	-0.0354	-0.4934	-1.9345	-3.8784	-5.7921	3.2428
DC-SCAN ($s_0 = Z_0 = 1$)	wire	-1.060136	-2.126863	-4.193878	-6.170789	-7.990438	+0.1532	+0.2734	+0.0338	-0.1061	-0.0886	0.1539
DC-SCAN ($s_0 = Z_0 = 1$)	sheet	-0.869388	-1.082998	-1.184779	-1.221408	-1.238506	+0.0954	+0.0963	+0.0606	+0.0311	+0.0147	0.0682
DC-SCAN refit ($s_0 = 0.986$, $Z_0 = 2.291$)	wire	-1.159757	-2.410988	-4.224040	-6.083858	-7.885874	+0.0536	-0.0107	+0.0036	-0.0192	+0.0160	0.0269
DC-SCAN refit ($s_0 = 0.986$, $Z_0 = 2.291$)	sheet	-0.927575	-1.087504	-1.181895	-1.220053	-1.237878	+0.0372	+0.0918	+0.0635	+0.0325	+0.0154	0.0550
PBE alone	wire	-1.225359	-3.865468	-20.817577	-109.470149	-545.596537	-0.0120	-1.4652	-16.5899	-103.4055	-537.6947	244.9839
PBE alone	sheet	-0.974775	-1.890956	-5.067933	-11.945449	-26.332795	-0.0100	-0.7117	-3.8225	-10.6929	-25.0796	12.3162
DC-PBE ($s_0 = Z_0 = 1$)	wire	-0.983007	-1.976722	-3.992240	-5.965216	-7.788276	+0.2303	+0.4236	+0.2354	+0.0994	+0.1136	0.2493
DC-PBE ($s_0 = Z_0 = 1$)	sheet	-0.811207	-1.021854	-1.152074	-1.205856	-1.231250	+0.1536	+0.1574	+0.0933	+0.0467	+0.0220	0.1093
DC-PBE refit ($s_0 = 1.013$, $Z_0 = 2.622$)	wire	-1.113806	-2.420413	-4.219246	-6.074800	-7.896149	+0.0995	-0.0201	+0.0084	-0.0102	+0.0057	0.0459
DC-PBE refit ($s_0 = 1.013$, $Z_0 = 2.622$)	sheet	-0.890883	-1.043359	-1.155006	-1.207227	-1.231886	+0.0739	+0.1359	+0.0904	+0.0453	+0.0213	0.0832

S6 Beyond the semilocal form: range separation, a quasi-1D LDA and local hybrids

All results below are fixed-density (fixed-orbital), exchange-only and exploratory; errors are $E_x - E_x^{\text{exact}}$ in mHa.

Range separation. With $\text{erf}(\omega r)/r = (2/\sqrt{\pi}) \int_0^\omega e^{-t^2 r^2} dt$, the exact long-range part is eq A44 with upper limit ω. At the HSE06 value $\omega = 0.11$ the short-range part carries 90% ($\lambda = 2$) to 98% ($\lambda = 10^4$) of the wire exchange, so replacing 25% of it by HF only scales the error: HSE06-type DC-SCAN (short-range DFA through the ITYH attenuation[S4]) gives +119, +220, +50, −50 and −36 for $\lambda = 2$, 10, 100, 10^3, 10^4. Long-range-corrected forms cannot remove the SCAN divergence, which comes from the short-range region near the axis, and the ITYH attenuation, calibrated on the 3D uniform gas, spoils the DC branches (LC-DC-SCAN with $\omega = 1$: −990 at $\lambda = 10^4$).

Quasi-1D LDA. For a product density $n_1(z) g_b(\rho)$ with a Gaussian transverse profile, the

exchange of the 1D uniform gas with the transverse-averaged interaction w_b is

$$e_x^{1D}(n_1) = -2\int_0^\infty \Big[\frac{\sin(kr)}{\pi r}\Big]^2 w_b(r)\,dr, \qquad w_b(x) = \frac{\sqrt{\pi}}{2b}\,\mathrm{erfcx}\Big(\frac{|x|}{2b}\Big), \qquad k = \frac{\pi n_1}{2}, \tag{S24}$$

and the local functional $E_x^{\mathrm{q1D}} = \int e_x^{1D}(n_1(z))\,dz$ uses $b = (2C_\perp)^{-1/2}$ from the density curvature (eq A16). On the Gaussian wire its errors are +90.3, +116.5, +120.2, +120.2, +120.2: the logarithm is exact and what remains is a λ-independent error of the axial inhomogeneity. Because the Kantorovich expression $E_x = -\frac{1}{4}\iint n_1(z)n_1(z')w_b(z-z')\,dz\,dz'$ is exact for every two-electron product density, the transferability of a correction can be tested on other axial profiles (Table S9). A 1D gradient term $-c\int n_1^2 h(s_1)\,dz$, $s_1 = |n_1'|/n_1^2$, $h = s^2/(1+s^2)$, $c = 0.1959$ fitted on the Gaussian, reaches 0.03 mHa on the Gaussian and on a three times wider Gaussian, but not on other shapes, and it cannot describe one orbital spread over two lobes, whose inter-lobe exchange is nonlocal.

Table S9: Quasi-1D LDA and the Gaussian-fitted 1D GGA on two-electron wires with different axial profiles $n_1(z)$ (normalized to 2), $\lambda = 10^3$. Computed exchange energies first (Ha; exact from the Kantorovich expression), then the errors of q1D-LDA and q1D-GGA (mHa).

	E_x (Ha)			error (mHa)	
$n_1(z) \propto$	exact, eq S20	q1D-LDA	q1D-GGA	q1D-LDA	q1D-GGA
e^{-z^2} (fit)	−6.06465	−5.94441	−6.06462	+120.2	+0.03
$e^{-z^2/9}$	−2.31374	−2.27366	−2.31373	+40.1	+0.01
$\mathrm{sech}^2 z$	−5.17175	−5.06758	−5.17851	+104.2	−6.75
e^{-z^4}	−6.91797	−6.81670	−6.90579	+101.3	+12.17
$(1+z^2)^{-2}$	−5.99886	−5.87876	−6.02237	+120.1	−23.51
$e^{-(z-1.5)^2} + e^{-(z+1.5)^2}$	−3.25469	−3.28592	−3.37617	−31.2	−121.48
$z^2 e^{-z^2}$	−4.54849	−4.63594	−4.76564	−87.5	−217.16

Local hybrids and $N > 2$. The local hybrid $E_x^{\mathrm{LH}} = \int [a\,e_x^{\mathrm{HF}} + (1-a)\,e_x^{\mathrm{DFA}}]\,d^3r$ with the iso-orbital weight $a = \tau_W/\tau$[S5] is exact for every one-orbital density ($a \equiv 1$). To test it beyond $N = 2$, the $N = 4$ and $N = 6$ wires occupy $\phi_k = g(\rho)h_k(z)$, $k < N/2$, with g the transverse Gaussian and h_k the Hermite functions, i.e. the s, p_z and d_{zz} functions of the aniGTO trap basis with $\boldsymbol{\omega} = (\lambda^2, \lambda^2, 1)$. The conventional exchange-energy density $e_x^{\mathrm{HF}}(\mathbf{r}) = -\sum_{ij}\phi_i\phi_j(\mathbf{r})\int \phi_i\phi_j(\mathbf{r}')|\mathbf{r}-\mathbf{r}'|^{-1}d^3r'$ is evaluated with the Gaussian transform of

$1/r$ (eq A43), which for these anisotropic Gaussian products reduces to a one-dimensional t integral with closed-form Gaussian moments; it integrates to the Kantorovich exact exchange to 3×10^{-7} Ha. The DFA energy densities come from libxc on a cylindrical logarithmic grid; DC-SCAN uses the full $F_x^{\mathrm{SCAN}}(s, \alpha)$ (eqs A38–A42) and reproduces Table 1 for $N = 2$ to all printed digits. Table S10 shows that the local hybrids converge to the exact exchange as λ grows, because the transverse kinetic energy, identical in τ and τ_W, dominates and $\langle 1 - a \rangle_n$ falls roughly as λ^{-2}. At $\lambda \leq 100$ the errors are still 17–250 mHa. DC-SCAN alone, whose 1D branch is fixed on $N = 2$, misses by -0.44 ($N = 4$) and -0.93 Ha ($N = 6$) at $\lambda = 10^4$.

Table S10: Exchange energies (Ha) and errors (mHa) of SCAN, DC-SCAN and of local hybrids ($a = \tau_W/\tau$) on N-electron Gaussian wires built from the aniGTO trap functions. For each N the computed values come first (exact, then every method), followed by the errors; $\langle 1 - a \rangle_n$ is the density-weighted mean of $1 - a$. For $N = 2$ every local hybrid is exact (0.000).

N		$\lambda = 2$	10	100	10^3	10^4
4	E_x^{exact}, eq S19 (Ha)	−2.714648	−5.825266	−10.828612	−15.879898	−20.932175
	SCAN E_x (Ha)	−2.793642	−8.222728	−37.068321	−161.286391	−670.385083
	DC-SCAN E_x (Ha)	−2.451291	−5.238404	−10.834946	−16.348922	−21.374615
	LH-PBE E_x (Ha)	−2.697921	−5.898669	−10.848097	−15.881711	−20.932300
	LH-SCAN E_x (Ha)	−2.761750	−5.928513	−10.850780	−15.881846	−20.932304
	LH-DC-SCAN E_x (Ha)	−2.737362	−5.911570	−10.847641	−15.881520	−20.932279
	$\langle 1 - a \rangle_n$	2.1×10^{-1}	2.6×10^{-2}	5.6×10^{-4}	8.6×10^{-6}	1.2×10^{-7}
	SCAN error (mHa)	−79.0	−2397.5	−26239.7	−145406.5	−649452.9
	DC-SCAN error (mHa)	+263.4	+586.9	−6.3	−469.0	−442.4
	LH-PBE error (mHa)	+16.7	−73.4	−19.5	−1.81	−0.125
	LH-SCAN error (mHa)	−47.1	−103.2	−22.2	−1.95	−0.129
	LH-DC-SCAN error (mHa)	−22.7	−86.3	−19.0	−1.62	−0.104
6	E_x^{exact}, eq S19 (Ha)	−4.348090	−9.820593	−18.912057	−28.124097	−37.338748
	SCAN E_x (Ha)	−4.439738	−13.222856	−59.804103	−261.133034	−1089.992571
	DC-SCAN E_x (Ha)	−3.982874	−8.856365	−18.916708	−29.066547	−38.266912
	LH-PBE E_x (Ha)	−4.305669	−9.991177	−18.965500	−28.129277	−37.339110
	LH-SCAN E_x (Ha)	−4.438678	−10.070617	−18.973239	−28.129682	−37.339123
	LH-DC-SCAN E_x (Ha)	−4.384159	−10.027305	−18.964633	−28.128757	−37.339051
	$\langle 1 - a \rangle_n$	3.1×10^{-1}	4.4×10^{-2}	9.9×10^{-4}	1.6×10^{-5}	2.1×10^{-7}
	SCAN error (mHa)	−91.6	−3402.3	−40892.0	−233008.9	−1052653.8
	DC-SCAN error (mHa)	+365.2	+964.2	−4.7	−942.5	−928.2
	LH-PBE error (mHa)	+42.4	−170.6	−53.4	−5.18	−0.362
	LH-SCAN error (mHa)	−90.6	−250.0	−61.2	−5.58	−0.375
	LH-DC-SCAN error (mHa)	−36.1	−206.7	−52.6	−4.66	−0.303

S7 Scope of the test

The coefficients follow from Gaussian reference profiles and are not exact constants of the exchange functional for arbitrary sheet or wire densities. The sheet and wire benchmarks use

one Gaussian profile each; three-dimensional norms (H, He, Ne, jellium), other confinement profiles and self-consistent tests remain to be done, and the refit of Table S8 has to be tested on those before it can be trusted; Hessian-level meta-GGAs have been implemented self-consistently elsewhere.[S6]

S8 Standalone codes

Every number and figure of the main text and this SI is produced by one of eight standalone Python scripts in the folder `codes/`. Each script is a single file that defines every function it uses (the DC-SCAN functional of Appendix A, the aniGTO integrals of Section S1 copied verbatim from aniGTO, the Kantorovich kernel and the SCE solver) and imports only numpy, scipy, pandas, matplotlib and, where libxc is needed, PySCF. A single command such as `python3 01_wire_benchmark.py` writes its results to `codes/results/` and its figures to `codes/figures/`.

Table S11: Standalone scripts, the manuscript items they produce, and typical laptop run times.

script	produces	time
`01_wire_benchmark.py`	Tables S1–S3, Figure S1	3 min
`02_sheet_benchmark.py`	Table 2, Tables S5, S6, Figure S2	2 min
`03_anigto_wire_libxc.py`	Table 1, Table S4	2 min
`04_exact_vs_proposed.py`	Table 4, Figure 1	5 min
`05_refit_switch_scales.py`	Table S8	1 min
`06_range_separation_q1d_lda.py`	range separation and quasi-1D LDA (Section S6)	$<$ 1 min
`07_q1d_gga_profiles.py`	Table S9	1 min
`08_local_hybrid_wires.py`	Table S10	2 min

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